Simultaneous Equations and Inequalities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises**

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 9 \Rightarrow y = 9$. Thus $x = 4(9) = 36$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

2 **[Foundational] • Target: Linear-Quadratic Simultaneous Systems**

A curve has equation $y = \frac{kx}{2x-1}$, where $xeq\frac{1}{2}$ and k is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that x is increasing at a rate of 0.6 units per second, find the rate of change of y when $x = 2$.

Step 1: Find the derivative $\frac{dy}{dx}$ in terms of k using the quotient rule. Let $u = kx$ and $v = 2x - 1$. Then $u' = k$ and $v' = 2$. $\frac{dy}{dx} = \frac{k(2x-1) - 2(kx)}{(2x-1)^2} = \frac{2kx - k - 2kx}{(2x-1)^2} = \frac{-k}{(2x-1)^2}$ Step 2: Find the parameter k. At $x = 1$, the gradient of the tangent is: $m_T = \frac{dy}{dx}\Big|_{x=1} = \frac{-k}{(2(1)-1)^2} = -k$ The gradient of the normal is given by $m_N = -\frac{1}{m_T} = \frac{1}{k}$. Since the gradient of the normal is.

3 **[Foundational] • Target: Quadratic Inequalities in One Variable**

Solve the logarithmic equation $\log_2(5x + 6) - \log_2(x) = 3$.

1. Apply the logarithm quotient law $\log_a(A) - \log_a(B) = \log_a\left(\frac{A}{B}\right)$: $\log_2\left(\frac{5x+6}{x}\right) = 3$ 2. Convert the equation from logarithmic to exponential form: $\frac{5x+6}{x} = 2^3 = 8$ 3. Multiply through by x (noting $x > 0$): $5x + 6 = 8x \Rightarrow 3x = 6 \Rightarrow x = 2$ 4. Checking the domain: Both $5(2) + 6 = 16 > 0$ and $2 > 0$, so $x = 2$ is a valid solution. Common misconceptions: - Calculating $2^3 = 6$ instead of 8 yields $5x + 6 = 6x \Rightarrow x = 6$..

4 **[Foundational] • Target: Rational Inequalities and Sign Analysis**

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.

To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$: Area $= \int_0^6 (6x - x^2)\,dx$ Integrate term by term: $\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$ Substitute the limits: $\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$ Common errors: - Estimating the region as a triangle with base 6 and maxim.

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5 [Foundational] • Target: Systems with Modulus and Geometric Applications

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.

To find the area enclosed by the curve and the x-axis, first find the x-intercepts by setting $y = 0$: $4x - x^2 = 0 \Rightarrow x(4 - x) = 0 \Rightarrow x = 0$ or $x = 4$. The area is given by the definite integral of y with respect to x from $x = 0$ to $x = 4$: $\int_0^4 (4x - x^2)\,dx = \left[2x^2 - \frac{x^3}{3}\right]_0^4$ Evaluating between the limits: $\left(2(4)^2 - \frac{4^3}{3}\right) - 0 = 32 - \frac{64}{3} = \frac{96 - 64}{3} = \frac{32}{3}$. Common misconceptio.

6 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

Take natural logs of both sides: $(x+2)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 2\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 2\ln 2 \Rightarrow x = \frac{2\ln 2}{\ln 5 - \ln 2}$. Final Answer: $x = \frac{2\ln 2}{\ln 5 - \ln 2}$

7 [Foundational] • Target: Linear-Quadratic Simultaneous Systems

The variables x and y are connected by the equation $y = \sqrt{2x + 5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

We are given $y = (2x + 5)^{\frac{1}{2}}$ and $\frac{dx}{dt} = 6$. Step 1: Differentiate y with respect to x using the chain rule: $\frac{dy}{dx} = \frac{1}{2}(2x + 5)^{-\frac{1}{2}} \cdot \frac{d}{dx}(2x + 5) = \frac{1}{2\sqrt{2x+5}} \cdot 2 = \frac{1}{\sqrt{2x+5}}$ Step 2: Evaluate $\frac{dy}{dx}$ at $x = 2$: $\left.\frac{dy}{dx}\right|_{x=2} = \frac{1}{\sqrt{2(2)+5}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$ Step 3: Use.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Quadratic Inequalities in One Variable

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants. Find the remainder when $P(x)$ is divided by $x - 3$.

Step 1: Use the Factor Theorem for the factor $x - 2$. Since $x - 2$ is a factor, $P(2) = 0$: $P(2) = 2(2)^3 + a(2)^2 + b(2) - 6 = 0$ $16 + 4a + 2b - 6 = 0 \Rightarrow 4a + 2b = -10 \Rightarrow 2a + b = -5$ $\quad - - -\ (1)$ Step 2: Use the Remainder Theorem for division by $x + 1$. Since the remainder is -24, $P(-1) = -24$: $P(-1) = 2(-1)^3 + a(-1)^2 + b(-1) - 6 = -24$ $-2 + a - b - 6 = -24 \Rightarrow a - b - 8 = -24 \Rightarrow a - b = -16$ $\quad - - -\ (2)$ Step 3: Solve the simultaneous equations.

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9 [Intermediate] • Target: Rational Inequalities and Sign Analysis

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

To form a committee of 5 members: 1. Committees with exactly 3 men: Since there are 5 members in total, the committee must contain 3 men and $5-3=2$ women. Number of ways $= \binom{n}{3} \times \binom{6}{2} = \binom{n}{3} \times \frac{6\times 5}{2} = 15\binom{n}{3}$ 2. Committees with exactly 4 men: The committee must contain 4 men and $5-4=1$ woman. Number of ways $= \binom{n}{4} \times \binom{6}{1} = 6\binom{n}{4}$ 3. Equating the two expressions: $15\binom{n}{3} = 6\binom{n}{4}$ Di.

10 [Intermediate] • Target: Systems with Modulus and Geometric Applications

In the expansion of $(1+ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

Using the binomial theorem, the expansion of $(1+ax)^n$ is given by: $(1+ax)^n = 1 + \binom{n}{1}(ax) + \binom{n}{2}(ax)^2 + \binom{n}{3}(ax)^3 + \ldots$ 1. Equate the coefficient of x: $\binom{n}{1}a = na = -12 \Rightarrow a = -\frac{12}{n}$ 2. Equate the coefficient of x^2: $\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 60 \Rightarrow n(n-1)a^2 = 120$ Substitute $a = -\frac{12}{n}$: $n(n-1)\left(-\frac{12}{n}\right)^2 = 120$.

11 [Intermediate] • Target: Simultaneous Equations & Inequalities Practice Exercises

Relative to a fixed origin O, ship A has an initial position vector of $(24\mathbf{i}+10\mathbf{j})$ km and moves with a constant velocity of $(-2\mathbf{i}+5\mathbf{j})$ km h^{-1}. Ship B has an initial position vector of $(-20\mathbf{i}-18\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i}+12\mathbf{j})$ km h^{-1}, where u is a constant. Given that ship B intercepts ship A, find the speed of ship B.

Let t be the time in hours after the ships begin moving. The position vector of ship A at time t is: $\mathbf{r}_A(t) = (24\mathbf{i}+10\mathbf{j}) + t(-2\mathbf{i}+5\mathbf{j}) = (24-2t)\mathbf{i} + (10+5t)\mathbf{j}$ The position vector of ship B at time t is: $\mathbf{r}_B(t) = (-20\mathbf{i}-18\mathbf{j}) + t(u\mathbf{i}+12\mathbf{j}) = (-20+ut)\mathbf{i} + (-18+12t)\mathbf{j}$ For ship B to intercept ship A, their positions must be identical at some time $t>0$, so.

12 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

1. Use the formula for the area of a triangle: Area $= \frac{1}{2}(AB)(AC)\sin(\angle BAC)$ $15\sqrt{3} = \frac{1}{2}(6)(10)\sin(\angle BAC) = 30\sin(\angle BAC)$ $\sin(\angle BAC) = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$ 2. Since $\angle BAC$ is obtuse ($90^\circ < \angle BAC < 180^\circ$): $\angle BAC = 180^\circ - 60^\circ = 120^\circ$ 3. Determine $\cos(\angle BAC)$: $\cos(120^\circ) = -\cos(60^\circ) = -\frac{1}{2}$ 4. Apply the cosine rule to find BC.

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13 [Intermediate] • Target: Quadratic Inequalities in One Variable

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ 1. Differentiate y with respect to x: $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$ 2. Evaluate $\frac{dy}{dx}$ at $x = 2$: $\frac{dy}{dx}\Big|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$ 3. Multiply by.

14 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Find the real solution to the equation $\log_3 x + \log_3 (x - 8) = 2$.

Apply the product rule of logarithms, $\log_a u + \log_a v = \log_a (uv)$: $\log_3 [x(x - 8)] = 2$ Convert the logarithmic equation to index (exponential) form: $x(x - 8) = 3^2$ $x^2 - 8x = 9$ $x^2 - 8x - 9 = 0$ Factor the quadratic: $(x - 9)(x + 1) = 0$ which gives potential solutions $x = 9$ or $x = -1$. Check the domain of the original logarithmic expression: for $\log_3 x$ and $\log_3 (x - 8)$ to be defined for real numbers, we must have $x > 0$ and $x - 8 > 0$, which requires $x > 8$. Thus, $x = -1$ is an extra.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

By the Remainder Theorem, when a polynomial $f(x)$ is divided by a linear factor of the form $ax - b$, the remainder is $f(\frac{b}{a})$. Here, the divisor is $2x - 1$, so we set $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$. Substitute $x = \frac{1}{2}$ into $f(x)$: $f(\frac{1}{2}) = 4(\frac{1}{2})^3 - 6(\frac{1}{2})^2 + 3(\frac{1}{2}) - 1 \ = 4(\frac{1}{8}) - 6(\frac{1}{4}) + \frac{3}{2} - 1$.

16 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The position vectors of points A and B relative to an origin O are given by $\vec{OA} = 5\mathbf{i} - 2\mathbf{j}$ and $\vec{OB} = -\mathbf{i} + 6\mathbf{j}$. Find the magnitude of $\vec{AB}$.

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (-\mathbf{i} + 6\mathbf{j}) - (5\mathbf{i} - 2\mathbf{j}) = (-1 - 5)\mathbf{i} + (6 - (-2))\mathbf{j} = -6\mathbf{i} + 8\mathbf{j}$ Next, calculate the magnitude $|\vec{AB}|$: $|\vec{AB}| = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ Common misconceptions: - Calculating $|\vec{OA} + \vec{OB}|$ gives $|4\mathbf{i} + 4\mathbf{j}| = \sqrt{16 + 16} = 4\sqrt{2}$. - Making a sign error when sub.

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17 [Challenging] • Target: Linear-Quadratic Simultaneous Systems • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the value of the integer $n \geq 3$ that satisfies the equation ${}^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$

Using the algebraic definitions of permutations and combinations: ${}^{n}\mathrm{P}_{3} = \frac{n!}{(n-3)!} = n(n-1)(n-2)$ ${}^{n}\mathrm{C}_{2} = \frac{n!}{2!\,(n-2)!} = \frac{n(n-1)}{2}$ Substitute these into the given equation: $n(n-1)(n-2) = 10 \times \frac{n(n-1)}{2}$ $n(n-1)(n-2) = 5n(n-1)$ Since $n \geq 3$, we have $n(n-1)e0$. Dividing both sides by $n(n-1)$ gives: $n-2 = 5 \Rightarrow n = 7$ Common misconceptions: - Forgetting the $2!$ denominator in ${}^{n}\mathrm{C}_{2}$ (treat.

18 [Challenging] • Target: Quadratic Inequalities in One Variable • Cross-Lesson: 6.01 Permutations and Combinations

Given that ${}^{n}P_{2} = n3$, where n is an integer such that $n \geq 3$, find the value of n.

Using the algebraic definitions of permutations and combinations: ${}^{n}\mathrm{P}_{2} = \frac{n!}{(n-2)!} = n(n-1)$ and $\binom{n}{3} = \frac{n!}{3!\,(n-3)!} = \frac{n(n-1)(n-2)}{3!} = \frac{n(n-1)(n-2)}{6}$ Setting them equal gives: $n(n-1) = \frac{n(n-1)(n-2)}{6}$ Since $n \geq 3$, $n(n-1)e0$. Dividing both sides by $n(n-1)$: $1 = \frac{n-2}{6}$ Multiplying by 6: $6 = n-2 \Rightarrow n = 8$ Common misconceptions: - Confusing $3!$ with 3 leads to $n-2 = 3 \Rightarrow n = 5$. - Forgetting the division by $3!$.

19 [Challenging] • Target: Rational Inequalities and Sign Analysis • Cross-Lesson: 7.01 Differentiation and Rates of Change

When the polynomial $\mathrm{P}(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x-3$, the remainder is 10. What is the value of the constant a?

According to the Remainder Theorem, when a polynomial $\mathrm{P}(x)$ is divided by $x-c$, the remainder is equal to $\mathrm{P}(c)$. Here, the divisor is $x-3$, so $c=3$. We evaluate $\mathrm{P}(3)$: $\mathrm{P}(3) = 2(3)^3 - 3(3)^2 + a(3) - 8$ $\mathrm{P}(3) = 2(27) - 3(9) + 3a - 8 = 54 - 27 + 3a - 8 = 19 + 3a$ Given that the remainder is 10: $19 + 3a = 10$ $3a = 10 - 19 = -9$ $a = -3$ Common misconceptions: - Choosing 3 results from a sign error when rearranging $19 + 3a = 10$ to $3a = 19 - 10 = 9$..

20 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 8.01 Integration and Kinematics

The position vectors of points A and B relative to a fixed origin O are given by $\vec{OA} = -2\mathbf{i} + 5\mathbf{j}$ and $\vec{OB} = 4\mathbf{i} - 3\mathbf{j}$. Find the magnitude of the vector $\vec{AB}$.

To find the vector $\vec{AB}$, subtract the position vector of A from the position vector of B: $\vec{AB} = \vec{OB} - \vec{OA} = (4\mathbf{i} - 3\mathbf{j}) - (-2\mathbf{i} + 5\mathbf{j}) = (4 - (-2))\mathbf{i} + (-3 - 5)\mathbf{j} = 6\mathbf{i} - 8\mathbf{j}$. The magnitude of $\vec{AB}$ is calculated using the Pythagorean theorem: $|\vec{AB}| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. Common misconceptions include adding the position vectors rather than subtracting them, giving.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|2x + 3| = 7$.

a. $x = 3$ and $x = -3$

b. $x = -2$ and $x = -\frac{10}{2}$

c. $x = 2$ only

d. **$x = 2$ and $x = -\frac{10}{2}$** ✓

Case 1: $2x + 3 = 7 \implies 2x = 4 \implies x = 2$. Case 2: $2x + 3 = -7 \implies 2x = -10 \implies x = -\frac{10}{2}$.

22 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|3x + 3| = 9$.

a. **$x = 2$ and $x = -\frac{12}{3}$** ✓

b. $x = 3$ and $x = -3$

c. $x = -2$ and $x = -\frac{12}{3}$

d. $x = 2$ only

Case 1: $3x + 3 = 9 \implies 3x = 6 \implies x = 2$. Case 2: $3x + 3 = -9 \implies 3x = -12 \implies x = -\frac{12}{3}$.

23 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|4x + 3| = 11$.

a. $x = 2$ only

b. $x = 3$ and $x = -3$

c. **$x = 2$ and $x = -\frac{14}{4}$** ✓

d. $x = -2$ and $x = -\frac{14}{4}$

Case 1: $4x + 3 = 11 \implies 4x = 8 \implies x = 2$. Case 2: $4x + 3 = -11 \implies 4x = -14 \implies x = -\frac{14}{4}$.

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24 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|5x + 3| = 13$.

a. $x = 2$ and $x = -\frac{16}{5}$ ✓

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{16}{5}$

d. $x = 3$ and $x = -3$

Case 1: $5x + 3 = 13 \implies 5x = 10 \implies x = 2$. Case 2: $5x + 3 = -13 \implies 5x = -16 \implies x = -\frac{16}{5}$.

25 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|6x + 3| = 15$.

a. $x = 3$ and $x = -3$

b. $x = 2$ and $x = -\frac{18}{6}$ ✓

c. $x = 2$ only

d. $x = -2$ and $x = -\frac{18}{6}$

Case 1: $6x + 3 = 15 \implies 6x = 12 \implies x = 2$. Case 2: $6x + 3 = -15 \implies 6x = -18 \implies x = -\frac{18}{6}$.

26 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation $|7x + 3| = 17$.

a. $x = 2$ and $x = -\frac{20}{7}$ ✓

b. $x = 2$ only

c. $x = -2$ and $x = -\frac{20}{7}$

d. $x = 3$ and $x = -3$

Case 1: $7x + 3 = 17 \implies 7x = 14 \implies x = 2$. Case 2: $7x + 3 = -17 \implies 7x = -20 \implies x = -\frac{20}{7}$.

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27 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|8x + 3| = 19 \).

a. **\(x = 2 \) and \(x = -\frac{22}{8} \)** ✓

b. \(x = 2 \) only

c. \(x = -2 \) and \(x = -\frac{22}{8} \)

d. \(x = 3 \) and \(x = -3 \)

Case 1: \(8x + 3 = 19 \implies 8x = 16 \implies x = 2 \). Case 2: \(8x + 3 = -19 \implies 8x = -22 \implies x = -\frac{22}{8} \).

28 [Foundational] • Target: Simultaneous Equations & Inequalities Practice Exercises

Solve the modulus equation \(|9x + 3| = 21 \).

a. \(x = 2 \) only

b. **\(x = 2 \) and \(x = -\frac{24}{9} \)** ✓

c. \(x = 3 \) and \(x = -3 \)

d. \(x = -2 \) and \(x = -\frac{24}{9} \)

Case 1: \(9x + 3 = 21 \implies 9x = 18 \implies x = 2 \). Case 2: \(9x + 3 = -21 \implies 9x = -24 \implies x = -\frac{24}{9} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Solve the simultaneous system $y - 2x = 3$ and $x^2 + y^2 = 26$ for the positive solution pair (x, y).

a. $(1, 5)$ ✓

b. $(2, 7)$

c. $(3, 9)$

d. $(4, 11)$

Substitute $y = 2x + 3$: $x^2 + (2x + 3)^2 = 26 \Rightarrow 5x^2 + 12x - 17 = 0 \Rightarrow (5x + 17)(x - 1) = 0 \Rightarrow x = 1, y = 5$.

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30 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Find the two points of intersection of the line $2x + y = 7$ and the rectangular hyperbola $xy = 6$.

a. $(2, 3)$ **and** $\left(\frac{3}{2}, 4\right)$ ✓

b. $(1, 5)$ and $(3, 1)$

c. $(2, 3)$ and $(3, 2)$

d. $(4, -1)$ and $(1, 6)$

$x(7 - 2x) = 6 \Rightarrow 2x^2 - 7x + 6 = 0 \Rightarrow (2x - 3)(x - 2) = 0 \Rightarrow (2, 3)$ and $(3/2, 4)$.

31 [Intermediate] • Target: Linear-Quadratic Simultaneous Systems

Determine the value of k for which the line $x + y = k$ is tangent to the curve $y = \frac{4}{x}$.

a. $k = \pm 4$ ✓

b. $k = 4$ only

c. $k = \pm 2$

d. $k = 8$

$x(k - x) = 4 \Rightarrow x^2 - kx + 4 = 0$. $\Delta = k^2 - 16 = 0 \Rightarrow k = \pm 4$.

32 [Intermediate] • Target: Quadratic Inequalities in One Variable

Find the set of values of x satisfying the quadratic inequality $2x^2 - 5x - 3 \geq 0$.

a. $x \leq -\frac{1}{2}$ **or** $x \geq 3$ ✓

b. $-\frac{1}{2} \leq x \leq 3$

c. $x < -1$ or $x > 3$

d. $x \geq 3$ only

$(2x + 1)(x - 3) \geq 0$. Parabola opens upward: $x \leq -1/2$ or $x \geq 3$.

33 [Intermediate] • Target: Quadratic Inequalities in One Variable

Solve the strict quadratic inequality $x(x + 4) < 12$.

a. $-6 < x < 2$ ✓

b. $x < -6$ or $x > 2$

c. $-2 < x < 6$

d. $0 < x < 3$

$x^2 + 4x - 12 < 0 \Rightarrow (x + 6)(x - 2) < 0 \Rightarrow -6 < x < 2$.

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34 [Intermediate] • Target: Quadratic Inequalities in One Variable

Find the range of values of x for which the expression $9 - x^2$ is strictly positive.

a. $-3 < x < 3$ ✓

b. $x < -3$ or $x > 3$

c. $x < 3$

d. $0 < x < 9$

$9 - x^2 > 0 \Leftrightarrow x^2 < 9 \Leftrightarrow -3 < x < 3$.

35 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Solve the rational inequality $\frac{x-2}{x+3} \leq 0$.

a. $-3 < x \leq 2$ ✓

b. $-3 \leq x \leq 2$

c. $x < -3$ or $x \geq 2$

d. $x \leq 2$

Sign changes at $x = -3$ and $x = 2$. Denominator cannot be 0, so $x \in (-3, 2]$.

36 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Find the complete solution set for the inequality $\frac{2x+1}{x-1} > 1$.

a. $x < -2$ **or** $x > 1$ ✓

b. $-2 < x < 1$

c. $x > 1$ only

d. $x > -2$

$\frac{2x+1-(x-1)}{x-1} = \frac{x+2}{x-1} > 0 \Rightarrow x < -2$ or $x > 1$.

37 [Intermediate] • Target: Rational Inequalities and Sign Analysis

Solve the inequality $\frac{1}{x} < 2$ for all non-zero real x.

a. $x < 0$ **or** $x > \frac{1}{2}$ ✓

b. $0 < x < \frac{1}{2}$

c. $x > \frac{1}{2}$ only

d. $x < \frac{1}{2}$

Multiply by $x^2 > 0$: $x < 2x^2 \Rightarrow x(2x - 1) > 0 \Rightarrow x < 0$ or $x > 1/2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 6.01 Permutations and Combinations

Find the values of x satisfying the simultaneous conditions $|x-3| \le 4$ and $x^2 - 4x > 0$.

a. $-1 \le x < 0$ **or** $4 < x \le 7$ ✓

b. $-1 \le x \le 7$

c. $0 < x < 4$

d. $4 < x \le 7$ only

$|x-3| \le 4 \Rightarrow [-1, 7]$. $x(x-4) > 0 \Rightarrow x < 0$ or $x > 4$. Intersection: $[-1, 0) \cup (4, 7]$.

39 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 7.01 Differentiation and Rates of Change

The perimeter of a right-angled triangle is 30 cm and its hypotenuse is 13 cm. Find the lengths of the two legs.

a. 5 cm **and** 12 cm ✓

b. 6 cm and 11 cm

c. 7 cm and 10 cm

d. 8 cm and 9 cm

$a + b = 17$, $a^2 + b^2 = 169 \Rightarrow 2ab = 17^2 - 169 = 120 \Rightarrow ab = 60$. Legs are 5 and 12.

40 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 8.01 Integration and Kinematics

Find the range of values of x where the line $y = 3x - 2$ lies entirely above the parabola $y = x^2 - 2x + 4$.

a. $2 < x < 3$ ✓

b. $x < 2$ or $x > 3$

c. $-3 < x < -2$

d. No such interval exists

$3x - 2 > x^2 - 2x + 4 \Rightarrow x^2 - 5x + 6 < 0 \Rightarrow (x-2)(x-3) < 0 \Rightarrow 2 < x < 3$.

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41 [Challenging] • Target: Simultaneous Equations & Inequalities Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. **c = 3, k = 4 ✓**

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

42 [Challenging] • Target: Linear-Quadratic Simultaneous Systems • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

a. (1, 2)

b. **(2, 1) ✓**

c. (3, 0)

d. (0, 3)

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

43 [Challenging] • Target: Quadratic Inequalities in One Variable • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Solve the inequality $x^2 + 9x + 20 < 0$.

a. $x - 4$

b. $-5 \leq x \leq -4$

c. **-5<x<-4 ✓**

d. $x4$

The quadratic factorises as $(x + 4)(x + 5)$ with roots -5 and -4; since the parabola opens upward, the expression is < 0 between the roots, giving $-5 < x < -4$.

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44 [Challenging] • Target: Rational Inequalities and Sign Analysis • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Solve the inequality $x^2 + 4x + 3 < 0$.

a. $x1$

b. **-3<x<-1 ✓**

c. $-3 \leq x \leq -1$

d. $x - 1$

The quadratic factorises as $(x + 3)(x + 1)$ with roots -3 and -1; since the parabola opens upward, the expression is < 0 between the roots, giving $-3 < x < -1$.

45 [Challenging] • Target: Systems with Modulus and Geometric Applications • Cross-Lesson: 6.01 Permutations and Combinations

Solve the inequality $x^2 - x - 6 < 0$.

a. $-2 \leq x \leq 3$

b. $x - 3$

c. **-2<x<3 ✓**

d. $x3$

The quadratic factorises as $(x - 3)(x + 2)$ with roots -2 and 3; since the parabola opens upward, the expression is < 0 between the roots, giving $-2 < x < 3$.

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